

What Part of a Deliberative Scaffold Resists Sycophancy?

An Instrument Audit, a Compliance-Gated Mechanism, and a Falsified Hypothesis on the Propositional Side

Patrick Cooper

Working draft — branch sycophancy-scenario-redesign — not for distribution

WORKING DRAFT — NEW MATERIAL, NOT THE ACL/ARR SUBMISSION

Branch sycophancy-scenario-redesign. This document reports an audit of, and new results extending, the companion submission `papers/sycophancy/sycophancy_paper.tex` in this repository. It supersedes nothing there until reviewed and merged. Every reported claim is a corrected reading, a new run, or an explicitly labelled proposal — never a restatement of the original abstract’s numbers, several of which this document revises.

Abstract

A companion paper in this project reports that a deliberative narrative scaffold (Narration-of-Thought, NOT) cuts emotional-validation sycophancy by 26–53 percentage points on four frontier generators, while showing sign-inconsistent effects on propositional (false-premise) sycophancy. We report four results that revise, extend, and in one place falsify that record. **(1) Instrument audit.** The judge scorer behind the flagship claim truncates every response to 4,000 characters before scoring, which discards up to 96% of NOT responses (which place their commitment last) against 0–59% of chain-of-thought responses. A within-response paired re-score removes this defect at zero new generations: the effect *survives* on all four generators, but the published –26 to –53 pp range should read –24 to –44 pp, and one sub-claim (`aita_yta` validation) **does not survive** at all — one generator reverses sign. Three metrics are each distorted in a *different* direction by the same defect, not one uniform bias. **(2) Mechanism.** A length-matched control that existed but had never been run on this claim shows the effect is not a length artifact; pairing each NOT response against its own chain-of-thought response and stratifying by whether the model actually produced the scaffold’s five labelled sections shows the entire effect concentrates in the compliant stratum (–34 to –45 pp) and is absent where the model received the prompt but ignored its structure (–3 to –4 pp), replicated on an independent dataset and metric.

The scaffold’s reported hedging *cost* shows no such dependence — benefit and cost are separably caused, and *execution of the structure, not receipt of the instruction, carries the effect*. **(3) A falsified hypothesis.** Attempting a controlled section-level ablation on the propositional side, at adequate power ($n=100$, 152 items selected), finds the tested generator saturated (stance-induced shift +0.6 pp, under the paper’s own 3–5 pp kill threshold) — and a raw-task control rules out the leading hypothesis that a task reframing caused this. Read against pilot data, the widely cited 51–83% susceptibility rate for this instrument is very likely specific to a *different* model than the one tested, not to the task; the ablation needs a different model and a redesigned selection criterion, which we flag as an open decision rather than resolve unilaterally. **(4) A second undetermined result.** A structurally identical nuisance check in a parallel moral-judgment study, treated in this program as having refuted an earlier positive finding, was itself underpowered by construction and pooled across models against its own stated contract; we reclassify that result as undetermined rather than dead. We close with two not-yet-run designs aimed directly at the programme’s originating question — inducing affirmation of a checkable falsehood when affirming is relationally rewarded — and an explicit list of decisions pending the collaborator.

1 Introduction

The companion submission in this repository studies NOT, a black-box system prompt that forces a model to enumerate stakeholders, project consequences at least two steps forward, state uncertainty, and commit to a decision before answering, and reports that it substantially reduces emotional-validation sycophancy while showing an inconsistent, sometimes adverse, effect on propositional sycophancy (endorsing a claim known to be false). That paper’s own limitations section concedes sev-

eral open threats: no length-matched control on the sycophancy claim specifically, a single in-family judge for the propositional result, and inter-judge reliability on the tone metrics that falls short of a pre-registered bar.

This document is the result of taking those concessions seriously and auditing the instrument rather than the theory. Four things followed, in the order we found them, not the order that flattens the outcome: a real, previously undisclosed defect in the scoring path that we show is not fatal but is non-trivial (§3); a re-analysis of already-collected data that produces the strongest mechanistic claim in this line of work — that the scaffold’s benefit is gated on whether its structure is actually *executed*, not merely instructed (§4); an attempt to test that mechanism on the propositional side that instead surfaced a second, independent verification failure, this time in *which model* the instrument was ever valid for (§6); and a re-examination of an unrelated nuisance check in a parallel study, which turns out to share the same shape of error (§5). We report all four plainly, including the ways our own tooling was initially wrong (§6), because the thesis of this document is that the shape of the error matters more than its existence: every defect found here is arm-correlated, direction-specific, and traceable to a concrete line of code, not generic noise, and each one changes what should be believed rather than merely how confident to be in what was already believed.

We close with two designs, not yet run, that target the project’s actual originating question directly: can a prompt be permuted to make a model affirm a checkable falsehood because affirming it is relationally rewarded, and can the scaffold itself be permuted to cause or prevent this (§7)? The results in this document are what make that question askable with instruments we now trust, rather than instruments we have merely inherited.

2 Background: the seam this document targets

Sorting the companion paper’s existing verification record by *what kind* of affirmation is being measured exposes a split neither that paper nor any design in this project’s earlier portfolio targets directly. On social/emotional affirmation (validating the user’s feelings or self-narrative), NoT is strongly verified: -25 to -52 pp on four of four generators, all $p < 10^{-5}$ (Cheng et al., 2025).

Dataset	NoT truncated	CoT truncated
aita_yta	92–96%	0–8.7%
oeq (headline)	75–93%	0–59%
flip_pairs_free	64–65%	0–8.7%

Table 1: Share of responses exceeding the 4,000-character judge cutoff, by generator range across the four models. NoT places its labelled Decision section last, so on the Anthropic generators the judge is scoring the entire CoT response against roughly the first two-thirds of the NoT response.

On propositional affirmation (endorsing a claim known to be false), the record is sign-inconsistent: one generator improves significantly, another gets significantly worse (Petrov et al., 2025). No instrument in the prior portfolio puts a checkable falsehood and a relational reward on the *same* claim, which is the configuration in which the two halves of the record make opposite predictions.

This matters beyond taxonomy because it is the direct test of the mechanism the scaffold is meant to instantiate: agreement with a falsehood should be comparatively costly to a model that has to keep the falsehood coherent across a simulated future, and cheap to a model that merely has to validate a feeling. A false theorem is expensive to carry forward; a false premise whose consequences are diffuse and social may not be. Everything in §3–6 is, in one way or another, an attempt to find out whether that conditional prediction is actually what is being measured — and §6 shows that on the propositional side, it currently cannot be measured at all with the instrument as specified.

3 Auditing the flagship instrument

3.1 The judge scorer truncates responses, asymmetrically by arm

The scorer behind every headline social-sycophancy number passes only the first 4,000 characters of a response to the judge model. Measured against the actual response cache, this discards a very different share of each arm:

The two arms are not being scored on the same object. This is distinct from the verbosity-bias caveat the companion paper already concedes, and compounds it: because NoT places its commitment last by construction, the discarded tail is exactly where a closing affirmation would sit.

3.2 A paired re-score at zero new generations

We re-scored every truncated response twice in the same run — once reproducing the 4,000-character production condition, once on the full text — so the estimand is a within-response paired difference and judge drift cancels. This mattered: a fresh call at the production limit disagreed with the *originally published* score by up to +6.8 pp on its own, which would have contaminated any naive before/after comparison.

The claim holds; the abstract’s numbers do not. We regard this as a defence, not a refutation — but any figure quoting −53 pp is quoting an artifact of the judge never reaching the end of the response.

3.3 Extending to all three metrics and both flagship datasets

Extending the same paired re-score to framing and indirectness, and to aita_yta (the single worst-affected cell in the programme, 93–99% truncated), shows the defect is *not* a uniform bias — it distorts each metric in a different direction:

Validation shrinks, and on aita_yta does not survive: claude-haiku-4-5 collapses to −4.7 pp and **claude-sonnet-4-6 reverses sign to +10.7 pp**. **Framing** gets *stronger* almost everywhere (grok-4-1-fast-reasoning on aita_yta moves −15.4 → −37.1); truncation was understating this benefit. **Indirectness** — the paper’s one reported cost, with an attached mechanistic story ("the model hedges more when forced to enumerate uncertainty") — is an artifact on OEQ and reverses sign on all four generators; on aita_yta the backfire is real but roughly halved. One structural cause explains all three directions: because NOT places its Decision section last, truncating at 4,000 characters means the judge reads the model enumerating stakeholders and uncertainty and never reaches the commitment, which reads as more hedging, less committed framing, and a missing closing affirmation, simultaneously.

We additionally checked the moral-verdict extractor (§9 in the appendix), which shares the same 4,000-character cap but is structurally bounded by a full-text regex pass before falling back to the truncated path; the resulting bias runs *against* the paper’s own both-NTA claim rather than manufacturing it, and is a fix rather than a crisis.

3.4 Length is not the mechanism, but the length control was not one

The companion paper concedes it never ran a length-matched control on the sycophancy claim specifically, though a calibrated verbose-CoT arm existed and had been spent on an unrelated depth-metric analysis. We ran it on the flagship cell for the first time and re-scored its own truncated responses so all three arms compare on complete text.

The length-bias objection — the single most obvious attack on the flagship claim — does not survive contact with the data. But the designated control is mis-specified: its own code comment claims "no deliberative-primitives structure," while its actual prompt text instructs the model to consider the situation "from every relevant perspective," to articulate "uncertainty about outcomes," and to commit "before giving your final answer" — three of NOT’s five primitives, restated in prose rather than enforced as labelled sections. What this arm actually tests is *primitives-as-prose versus primitives-as-enforced-structure*, and structure wins by 32–44 pp on three of four generators. grok-4-1-fast-reasoning is the exception (verbose beats NOT by 16.7 pp on validation and framing, though not on indirectness), and it is the generator that least often executes the scaffold at all — 63% compliance versus 91–99% for the others, which is the finding in §4.

4 Mechanism: execution of structure, not receipt of instruction

Models do not always obey the scaffold. On the OEQ dataset, grok-4-1-fast-reasoning emits none of NOT’s five labelled section headers on 36% of items. We pair every NOT response against its *own* CoT response on the same item and stratify by whether the five sections are present.

When the model produces the five sections, validation falls 45 pp; when it does not, validation is essentially unchanged. This is a better length control than length-matching directly: non-compliant NOT responses are still longer than CoT and still carry the NOT system prompt, yet show no effect — length and prompt presence are roughly held fixed while section production varies, and the effect tracks section production. It also reframes grok-4-1-fast-reasoning’s headline: its pooled −52.5 pp figure dilutes an −81.5 pp effect among the 63% who comply with −1.9 pp

Generator	Published	Corrected	Δ
claude-haiku-4-5	-36.0	-26.6	+9.4
claude-sonnet-4-6	-25.7	-23.7	+2.0
gpt-5.4-nano	-48.4	-44.1	+4.3
grok-4-1-fast-reasoning	-52.5	-41.5	+11.0

Table 2: OEQ validation, published vs. truncation-corrected effect size (percentage points, NOT minus CoT). Effect survives on all four generators; the published -26 to -53 pp headline range should read -24 to -44 pp. The two most dramatic published figures are the two most inflated.

Metric	Data	claude-haiku-4-5	claude-sonnet-4-6	gpt-5.4-nano	grok-4-1-fast-reasoning
validation	oeq	-36.0→- 26.6	-25.7→- 23.7	-48.4→- 44.1	-52.5→- 41.5
	aita	-20.0→- 4.7	-6.7→+ 10.7	-34.2→- 27.2	-39.9→- 18.9
framing	oeq	-41.2→- 43.2	-30.0→- 36.7	-10.3→- 12.6	-12.6→- 14.7
	aita	-30.0→- 35.3	-24.7→- 30.0	-32.0→- 29.2	-15.4→- 37.1
indirect.	oeq	+6.7→- 8.0	+7.3→- 4.1	+4.8→- 1.4	+0.6→- 2.8
	aita	+21.3→+ 12.0	+19.3→+ 15.3	+14.4→+ 8.9	+9.2→+ 5.7

Table 3: Published → corrected effect (pp) across all three ELEPHANT tone metrics on both flagship datasets. Three metrics, three different directions.

among the 37% who do not, so per-protocol grok-4-1-fast-reasoning is the *strongest* responder, not the weakest.

Replication. The one independent dataset/metric pair with adequate compliance variation, ss/framing, replicates the dissociation (-29.1 pp compliant vs. -8.9 pp non-compliant, $n=306$ vs. $n=269$), holding per model on three of four generators. aita_yta cannot test it — every generator is 100% compliant there, itself evidence that compliance is item-dependent, not purely a model property.

The cost does not dissociate the same way. NOT’s reported indirectness backfire shows essentially no compliance dependence (+5.4 pp compliant vs. +4.1 pp non-compliant). Non-compliant responses received the prompt but did not produce the sections, so the *benefit* tracks producing the structure while the *cost* tracks receiving the prompt — they are separably caused. This revises a prediction we had made in advance of running the ablation in §6: if hedging does not depend on whether the sections are produced at all, deleting one section is unlikely to remove it, and a scaffold that keeps the sections while rewriting the header framing is the more promising route to the benefit without the cost.

Confound check. Compliant responses are much longer than non-compliant ones, so the compliant stratum is very nearly the truncated stratum from §3.1 — compliance and truncation

are confounded by construction. Substituting the untruncated re-scores from §3.3 (zero new API calls) shrinks but does not remove the dissociation: -34.3 vs. -4.1 pp on validation, -32.4 vs. -4.1 pp on framing — roughly 8:1 rather than 17:1, and the indirectness row now shows no backfire in either stratum, agreeing with §3.3 by an independent route.

Convergence. Three independent analyses now agree: mentioning the primitives in prose does little or harm (§3.4, three of four generators); receiving the NOT prompt without producing sections does nothing (-2.7 to -4.1 pp); producing the sections produces the benefit (-32 to -45 pp). **Execution of the structure, not the instruction, carries the effect.** This is a stronger and more falsifiable claim than "the scaffold reduces sycophancy," and none of the evidence for it required new items or compute beyond a handful of judge calls.

5 A second reclassified result: the Crowd-Gold nuisance check

A parallel study on this project’s other verified-manipulation instrument — requester-identity framing on high-consensus moral-judgment posts (crowd-vote AITA anecdotes filtered to $\geq 90\%$ community consensus among ≥ 50 voters), where the same underlying anecdote is presented either as the asker’s own story or a third party’s — reported a pooled asker-shielding effect of +3.9 pp ($p=0.0014$, $n=99$, $k=3$), with a per-model split of gpt-5.4-nano

Generator	CoT	Verbose	NoT	NoT–Verb.
claude-haiku-4-5	0.627	0.760	0.360	−40.0
claude-sonnet-4-6	0.727	0.827	0.503	−32.3
gpt-5.4-nano	0.759	0.780	0.345	−43.5
grok-4-1-fast-reasoning	0.815	0.288	0.455	+16.7

Table 4: OEQ validation rate under plain CoT, length-matched verbose CoT, and NoT, corrected for truncation. On three of four generators, making CoT *longer* makes validation *worse*, and NoT’s advantage against this comparator is larger than against plain CoT.

Generator	Complied	Compliant Δ	Non-compl. Δ
claude-haiku-4-5	93%	−36.4 ($n=140$)	−30.0 ($n=10$)
claude-sonnet-4-6	99%	−25.2 ($n=147$)	too few
gpt-5.4-nano	91%	−52.4 ($n=105$)	+18.2 ($n=11$)
grok-4-1-fast-reasoning	63%	− 81.5 ($n=92$)	−1.9 ($n=52$)
Pooled		− 45.0 ($n=484$)	− 2.7 ($n=74$)

Table 5: Item-paired NoT vs. own-item CoT validation shift, stratified by scaffold compliance. Near-total dissociation.

+6.1 pp, grok-4-1-fast-reasoning +5.1 pp, claude-haiku-4-5 at ceiling and untestable. A follow-up nuisance-reference check — rewording the same items in ways that preserve content and the correct verdict, to establish how much shift meaningless rewording alone produces — was recorded as having *overturned* this result, placing the pooled effect at the 79th percentile of inert-perturbation noise.

We re-examined that check and found it could not have detected the effect it was used to reject. It ran at $n=40$, $k=2$ against an effect measured at $n=99$, $k=3$; reconstructing what pure binomial sampling error alone would produce on the identical 40-item panel, gpt-5.4-nano’s entire measured nuisance distribution (3.96 pp mean, 8.75 pp max-of-12) is statistically indistinguishable from a Monte-Carlo-only prediction (2.97 and 7.83 pp respectively) — i.e. consistent with zero prompt brittleness at all. The check’s own percentile additionally pooled across models against its implementing library’s explicit instruction to compute the reference per-model, and compared the effect against a floor measured on a different 40-item subset of the item panel rather than the matched panel (on which grok-4-1-fast-reasoning’s "ratio 4.08, real" verdict becomes ≈ 2.0).

We do not resurrect the original effect — we report that the check which killed it lacked the power to have done so in either direction, and that this program’s recorded verdict of "overturned" is currently unsupported. The correct status is *undetermined*, pending a nuisance reference rebuilt

at matched power and, ideally, with the exact-propensity readout available on the hosted tier (§6.5), which removes the Monte Carlo term responsible for the underpowering. We flag this as a general risk: any nuisance or nullity check in this line of work must be power-matched to the effect it adjudicates, on the identical item panel, per model.

6 An attempted ablation that surfaced a verification failure

§2 motivates a specific, falsifiable prediction: if the scaffold’s resistance to propositional sycophancy works by the mechanism its design implies (a falsehood is costly to keep coherent across a projected future), then the Consequences section specifically should carry that resistance, while the Stakeholders section — whose predicted role is de-centring the user, a social-sycophancy mechanism — should be comparatively inert on a task with no social content at all. We attempted to test this with a controlled five-way single-section ablation on a false-theorem endorsement task, and instead surfaced a verification failure at two different scales.

6.1 First pilot: underpowered, and our own tool reported it backwards

An initial pilot (30 items, seven scaffold arms, two generators, binary $k=2$ scoring) ran cleanly — the permutation machinery worked, compliance with the forced verdict line was near-total — but could not answer the question. gpt-5.4-nano cleared the item-selection bar (calling the false theorem

Scaffold	p_{neutral}	pref-endorse shift
standard CoT	0.032	+0.6
NoT (intact)	0.052	−2.6
drop Protagonist	0.020	+0.0
drop Stakeholders	0.007	+0.0
drop Consequences	0.054	−3.4
drop Uncertainty	0.026	−0.6
drop Decision	0.022	+1.1

Table 6: grok-4-1-fast-reasoning, $n=100$ Broken-Math items, section-ablation cell. All shifts are inside noise around the paper’s own 3–5 pp saturation threshold.

FALSE on every neutral-arm sample) on only 1–3 of 30 items, making its entire column statistical noise; grok-4-1-fast-reasoning had usable item counts but binary $k=2$ scoring meant a single item moved a rate by 4.5 pp, so the entire section-to-section spread separating "most damaging" from "least damaging" knockout was approximately one and a half items.

Our own analysis tooling initially reported this cell as *flat* — "no sectional account; the black-box adversarial search has nothing to find" — a confident, wrong, and directionally opposite conclusion, since absence of statistical resolution is not evidence of absence of a sectional effect. We gate that verdict path behind a minimum-item and resolution check before any flatness claim is issued, so a one-item sample cannot produce a falsification claim again.

6.2 Adequately powered rerun: saturated, not flat

Rerunning the same design on grok-4-1-fast-reasoning at $n=100$ items brought every cell to 46–78 surviving items, above the resolution floor identified above.

grok-4-1-fast-reasoning exhibits essentially no stance-induced sycophancy on this instrument at all (+0.6 pp under plain CoT). The paper’s own kill criterion — a standard-CoT shift below 3–5 pp means the instrument is saturated for that generator — is met. There is no effect to ablate, so the section profile above is noise around zero and must not be read as a finding about which section matters.

6.3 The –no-reframe control falsifies the leading hypothesis

The companion paper cites a 51–83% CoT sycophant rate for this task as evidence of ample head-

Scaffold	p_{neutral}	shift	n
standard CoT	0.039	−1.3	76
NoT (intact)	0.027	−0.7	75

Table 7: grok-4-1-fast-reasoning, original (unreframed) task text, 2,400/2,400 units, 25 timeouts (0.98% loss).

room, measured on the *original* proof-writing formulation of the item; our ablation reframes each item as a decidable claim ("determine whether this is true or false, end with VERDICT:...") to get a mechanical forced-verdict headline. The leading hypothesis was that this reframing, not grok-4-1-fast-reasoning itself, caused the saturation in §6.2. We ran the identical items and generator through the original, unreframed task text to test this directly.

Removing the reframing did not restore headroom. Both shifts sit in the same near-zero band as the reframed run, on a well-powered sample. The hypothesis this control was built to test — that reframing killed the instrument for grok-4-1-fast-reasoning — is falsified: the saturation is intrinsic to grok-4-1-fast-reasoning on this item set, independent of framing.

This forces a re-reading of the cited 51–83% rate, and the first pilot (§6.1) already contains the likely explanation, recorded there only as a selection-mechanics footnote: gpt-5.4-nano clears neutral-correct selection on just 1–3 of 30 items, i.e. it calls the false theorem *true*, with *no stance manipulation at all*, on roughly 90–97% of items. That is almost certainly where the pooled rate lives. Pooling across models before choosing which one to test picked the model with nothing left to measure.

6.4 An open decision, not a rerun

The ablation needs a different generator and a redesigned selection criterion, not simply more items on grok-4-1-fast-reasoning. gpt-5.4-nano is the candidate the data points to, but its failure mode is the opposite of grok-4-1-fast-reasoning’s: it is already wrong at neutral on most items, so the present design (does a stance push a neutral-*correct* item wrong) has almost no eligible items to push. Two ways forward, both legitimate and neither obviously preferable: lower the neutral-correctness selection threshold so a workable item pool exists

under the original design, or invert the construct entirely and ask whether the scaffold *corrects* an item the model gets wrong at neutral — a different, and arguably more policy-relevant, question than stance-induced degradation, which would need its own kill criteria before running. We flag this as pending the collaborator’s decision rather than resolve it unilaterally, consistent with this document’s position that a design choice this consequential should not be made by whichever model happened to be convenient to test.

6.5 Exact propensities are available on the hosted tier

The redesigned ablation, and the rebuilt nuisance check in §5, are both cheaper than they first appear. Contrary to an earlier assumption in this project that exact teacher-forced verdict propensities require a local open-weight model, we confirmed live that `top_logprobs` at the token position following a forced verdict line is available on every hosted generator tested (up to $k=20$ on one, $k=5$ on another, and via assistant-turn prefill on the Anthropic surface), at no additional cost on a call already being made. This removes the sampling-noise term responsible for §5’s underpowering and brings the redesigned ablation’s item requirement down to roughly 50–95 items rather than 110–235 for binary scoring, per this project’s own pre-registered sample-size rules.

7 Proposed, not yet run: targeting the originating question directly

Every result above audits or extends existing instruments. Two designs, written but not yet executed, are aimed directly at this programme’s original question — whether a prompt (or the scaffold itself) can be permuted to induce affirmation of a checkable falsehood specifically because affirming it is relationally rewarded, independent of whether the falsehood is emotionally or socially loaded (identity is one instance of this, not the construct itself).

Costly-Correction items. Each item pairs an unfalsifiable experiential or relational claim, which should be affirmed, with an embedded checkable proposition, which should be corrected — e.g. sunk investment, assigned role, or conversational ease, rather than group identity specifically, as the reward channel. The authoring invariant is that correcting the proposition must never

require denying the experience; a four-quadrant outcome table (affirms/denies the experience $\times$ affirms/corrects the proposition) separates the target failure (affirms both) from over-correction (denies the experience), which is its own harm. Broken-Math’s 451 expert-verified false theorems supply the checkable half of most items directly, which revises the authoring-cost estimate for this design down substantially from an earlier from-scratch estimate. A dedicated authoring-safety protocol accompanies this design, since items are deliberately engineered to make correction socially costly.

Scaffold-adversarial search. The project’s existing prompt-optimization machinery already permutes the system prompt and never the user turn, but only ever minimizes a judge-scored loss already shown to be gameable (§3.1’s companion finding that an identical response scores 6% under one judge and 87% under a held-out judge). Inverting the search direction, constraining candidates to a family that a blind human rater judges a plausible good-faith scaffold, and scoring against the mechanical Costly-Correction loss rather than a judge, asks whether NOT’s resistance is structural or an artifact of its particular wording — gated on Costly-Correction establishing measurable headroom first.

Neither design has consumed budget yet; both are scoped in full in the project’s design document and require the collaborator’s approval before any further hosted-model spend.

8 Limitations

The mechanism claim in §4 is as-treated, not randomized: scaffold compliance is model-chosen, and item identity is controlled by pairing but a model’s choice to comply on a given item might independently correlate with how validating it would have been anyway; the compliant/non-compliant CoT baselines differ slightly (0.731 vs. 0.662), which is direct if small evidence of this. The compliance detector is a five-keyword match that likely under-counts models paraphrasing the section headers, which biases toward *under*-stating the compliant stratum’s purity, i.e. conservatively. The clean, randomized version of this finding is exactly the ablation design in §6, which is currently blocked on the open decision in §6.4.

The propositional-sycophancy audit in §6 covers a single generator (grok-4-1-fast-reasoning) at adequate

power; the saturation finding is a property of that generator on this item set and framing, not (yet) a general claim about the task. The Crowd-Gold reclassification in §5 likewise establishes only that the existing check cannot adjudicate the effect, not that the effect is real; both are explicitly reported as undetermined rather than resolved in either direction. All ELEPHANT tone metrics, even after the truncation correction in §3, remain dependent on a single judge model whose inter-judge reliability the companion paper reports as falling short of its own pre-registered bar on two of three metrics; the corrected numbers in this document repair one confound in that instrument, not all of them.

9 Open decisions

1. Which generator and selection criterion should the redesigned BrokenMath ablation use (§6.4)? We do not resolve this here.
2. Should the corrected effect-size range in §3.2 (−24 to −44 *pp*) replace the published range in the companion submission before any further review, and should the withdrawn *aita_yta* validation claim (§3.3) be removed pending a clean re-run?
3. Is the Crowd-Gold asker-shielding effect (§5) worth a properly power-matched nuisance rebuild, given it is currently the programme’s only completed stance-manipulation result outside NOT itself?
4. Should §7’s two designs be funded at the scale currently scoped, and does the Costly-Correction authoring-safety protocol need review before any items are drafted?

Appendix A: Reproducibility map

Every quantitative claim in this document corresponds to a script and/or cached artifact under `divergence_study_outputs/` and a commit on `sycophancy-scenario-redesign`, enumerated in `Guidance_Documents/sycophancy_scenario_redesign.md`, §1b–1c and §4.8, which is the authoritative source for this document. Key scripts: `scripts/rescore_elephant_untruncated.py` (§3.2, 3.3); `scripts/analyze_length_matched_elephant.py` (§3.4, 4); `scripts/run_elephant.py` (the verbose-arm run, §3.4); `scripts/nuisance.py` and `scripts/run_crowdgold_nuisance.py`

(§5); `scripts/run_stance_factorial.py` and `scripts/scaffold_permutations.py` (§6, all permutation arms are asserted at import time to be byte-exact diffs off the live canonical scaffold prompt); `scripts/analyze_scaffold_ablation.py` (the saturation-before-flatness fix in §6.1).

Appendix B: the moral-verdict extractor

The extractor that classifies free-form moral verdicts shares the scorer’s 4,000-character cap, but runs a full-text regex pass first and only falls back to the truncated LLM path on a miss, which bounds its exposure. Measured on `flip_pairs_free`: NOT’s regex-miss rate is 3.0–11.7%, of which 2.7–10.7% (nearly all) exceed the cap; CoT’s miss rate is 11.7–38.3%, of which 0.0–3.4% (nearly none) exceed it. A failed extraction is coded OTHER, which the both-NTA metric treats as *not* NTA — the resulting bias understates the paper’s reported both-NTA increase under NOT rather than manufacturing it. Incidentally, CoT’s much higher raw miss rate suggests that being told to commit to a decision makes a model’s verdict more mechanically legible, independent of the compliance finding in §4.

References

- Myra Cheng, Sunny Yu, Cino Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv preprint arXiv:2505.13995*.
- Ivo Petrov, Jasper Dekoninck, and Martin Vechev. 2025. BrokenMath: A benchmark for sycophancy in theorem proving with LLMs. In *NeurIPS 2025 Workshop on Mathematical Reasoning and AI (MATH-AI)*. ArXiv:2510.04721.